Contributors

Project Team

This project is built and maintained by the following contributors:

#	Name	Role
1	**Siddhant Udavant**	Project Lead & Core Architecture
2	**Trupti Shinde**	Matrix Math Engine & Algorithms
3	**Kanika Shinde**	UI/UX Design & Frontend Development
4	**Aayush Shewale**	Canvas Rendering & Image Processing
5	**Shruti Sonawane**	Testing, Documentation & Quality Assurance

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Contribution Details

1. Siddhant Udavant — Project Lead & Core Architecture

- Defined the overall system architecture and project structure.
- Designed the module separation (`matrix.js`, `transform.js`, `presets.js`, `ui.js`, `app.js`).
- Implemented the application entry point and initialization logic.
- Coordinated integration between all modules.
- Managed version control workflow and code reviews.
- Oversaw project planning, task assignment, and milestone tracking.

2. Trupti Shinde — Matrix Math Engine & Algorithms

- Implemented the 3×3 matrix math engine (`matrix.js`).
- Developed core linear algebra functions: `multiply()`, `invert()`, `determinant()`, `transpose()`, `identity()`.
- Implemented the `applyToPoint()` function with perspective division.
- Built the transformation composition (matrix chaining) logic.
- Developed preset transformation factory functions (`presets.js`): rotation, scaling, translation, shearing, reflection, and perspective matrices.
- Validated all matrix operations against known mathematical results.

3. Kanika Shinde — UI/UX Design & Frontend Development

- Designed the complete user interface layout and visual styling (`style.css`).
- Built the interactive 3×3 matrix grid editor with real-time input validation.
- Implemented drag-and-drop image upload functionality.
- Created preset transformation buttons and parameter sliders.

- Developed the undo/redo history UI and action bar controls.
- Implemented keyboard shortcuts (Ctrl+Z, Ctrl+Y, Ctrl+S, Ctrl+R, Ctrl+O).
- Ensured responsive design across desktop and tablet viewports.

4. Aayush Shewale — Canvas Rendering & Image Processing

- Implemented the HTML5 Canvas rendering pipeline for original and transformed views.
- Developed the pixel-level image transformation engine (`transform.js`).
- Built the inverse mapping algorithm to eliminate holes in transformed output.
- Implemented nearest-neighbor and bilinear interpolation modes.
- Handled boundary checking for out-of-bounds pixel lookups.
- Integrated native Canvas `setTransform()` for GPU-accelerated affine transforms.
- Implemented the image export (download) functionality in PNG/JPEG formats.

5. Shruti Sonawane — Testing, Documentation & Quality Assurance

- Authored the project `README.md` with detailed mathematical documentation.
- Created the `CONTRIBUTING.md` (this file) with contributor details.
- Performed cross-browser testing (Chrome, Firefox, Edge, Safari).
- Validated all transformation presets against expected mathematical outputs.
- Tested edge cases: large images, zero-determinant matrices, extreme parameter values.
- Documented keyboard shortcuts, UI controls, and usage instructions.
- Reviewed code for consistency, style adherence, and correctness.

Contribution Summary

Contributor	Modules Owned	Key Deliverables
Siddhant Udavant	`app.js`, Architecture	System design, integration, project management
Trupti Shinde	`matrix.js`, `presets.js`	Matrix operations, preset transforms, math validation

Contributor	Modules Owned	Key Deliverables
Kanika Shinde	`ui.js`, `style.css`	UI design, event handling, keyboard shortcuts, responsive layout
Aayush Shewale	`transform.js`, Canvas	Rendering pipeline, interpolation, image export
Shruti Sonawane	`README.md`, `CONTRIBUTING.md`	Documentation, testing, quality assurance

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Acknowledgments

We would like to thank the open-source community and the following resources that guided this project:

- **OpenCV** documentation for reference implementations of affine and perspective transforms.
- **GeeksforGeeks** articles on 2D transformation matrices.
- **MDN Web Docs** for Canvas 2D API reference.
- **Linear Algebra and Its Applications** by Gilbert Strang for mathematical foundations.

Built with collaboration, powered by linear algebra.